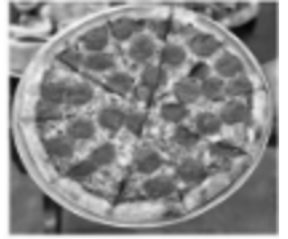


Geometry
Unit 10
Lesson 1 Practice

Name_____ **Answer Key** _____
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Use the given context to answer questions 1-10.

Eduardo orders a small circular pizza that has a 10-inch (in.) diameter and is cut into 6 triangular slices.



1. What is the approximate area of a 10-inch pizza? Use 3.14 for π .
 $A = 3.14(5)^2 = 3.14(25) = 78.5 \text{ in}^2$

2. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 3 slices?

Fraction: $\frac{1}{2}$

Percentage: 50%

3. What is the area of 3 slices?

$A = 39.25 \text{ in}^2$

4. What is the central angle of 3 slices?

180°

5. If the slices are cut precisely and equally, what fraction of the pizza is 5 slices?

$\frac{5}{6}$

6. What is the area of 5 slices?

$$A = 65.42 \text{ in}^2$$

7. What is the central angle of 5 slices?

$$300^\circ$$

8. If the slices are cut precisely and equally, what fraction of the pizza is 1 slice?

$$\frac{1}{6}$$

9. What is the area of 1 slice?

$$A = \frac{78.5}{6} = 13.08 \text{ in}^2$$

10. What is the central angle of 1 slice?

$$\frac{360}{6} = 60^\circ$$

Geometry
Unit 10
Lesson 2 Practice

Name Answer Key
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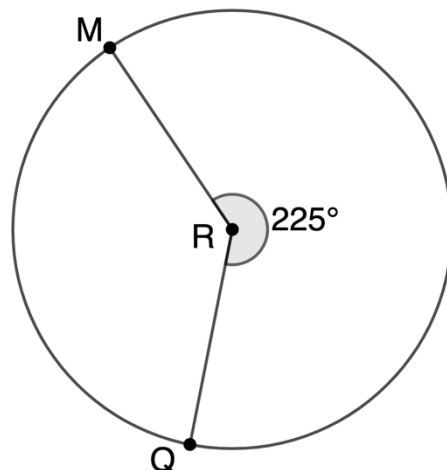
For questions 1-2, Circle R is shown where points M and Q are on the circle. The measure of central angle QRM is 225° .

1. What is the ratio of the central angle, $\angle QRM$, to the whole circle? Simplify the ratio.

$$\frac{225}{360} = \frac{5}{8}$$

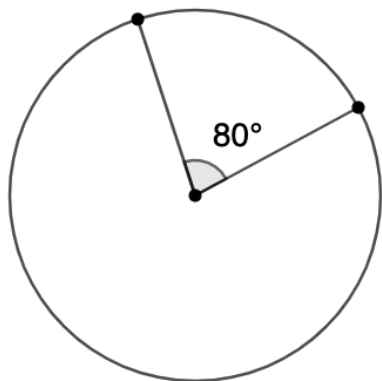
2. What fraction of the circle's circumference does $\widehat{MQ}$ represent?

$$\frac{225}{360} = \frac{5}{8} \text{ for the larger arc; } \frac{135}{360} = \frac{3}{8} \text{ for the smaller arc}$$



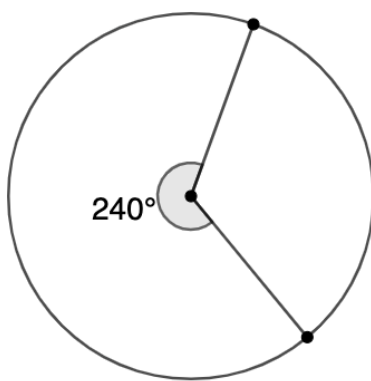
For questions 3-8, find the ratio of the arc length to the circumference of the circle.

3.



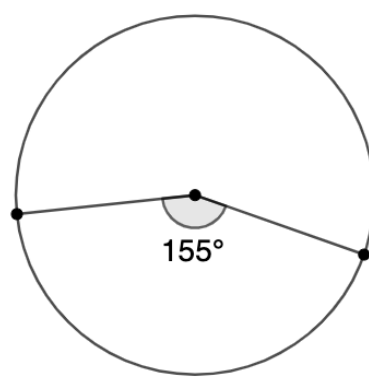
$$\frac{80}{360} = \frac{2}{9}$$

4.



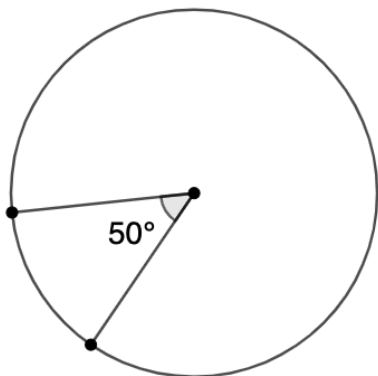
$$\frac{240}{360} = \frac{2}{3}$$

5.



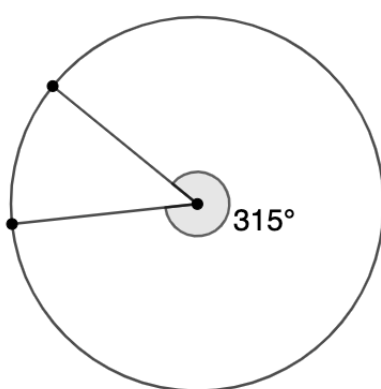
$$\frac{155}{360} = \frac{31}{72}$$

6.



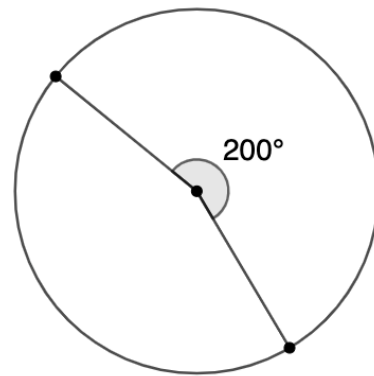
$$\frac{50}{360} = \frac{5}{36}$$

7.



$$\frac{315}{360} = \frac{7}{8}$$

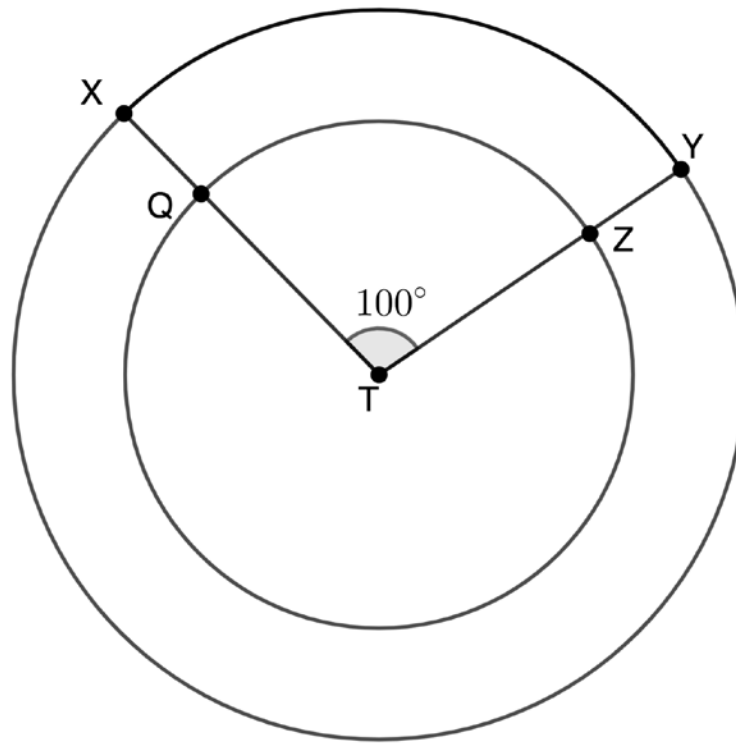
8.



$$\frac{200}{360} = \frac{5}{9}$$

For questions 9-10, consider the following information.

Points X and Y lie on Circle T . The radius $\overline{TX}$ measures 10 inches (in.). A smaller Circle T with a radius measure of 7 in. rests inside the larger Circle T , where points Q and Z lie on the smaller circle. The smaller Circle T has central angle $\angle QTZ$ that measures 100° and the larger Circle T has central angle $\angle XTY$ that measures 100° .



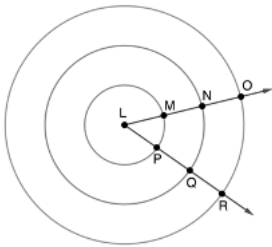
9. What is the ratio of the arc length to circumference of Circle T with radius measure of 10 in.?

$$\frac{100}{360} = \frac{5}{18}$$

10. What is the ratio of the arc length to circumference of Circle T with radius measure of 7 in.?

$$\frac{100}{360} = \frac{5}{18}$$

For questions 1-6, concentric circles with center L are shown. Points M and P lay on the innermost Circle L , and has radius $LM = 3$ units. Points N and Q lay on the middle Circle L , and has radius $LN = 6$ units. Points O and R lay on the outer Circle L , and has radius $LO = 9$ units.



Determine the circumference for each circle. Leave your answer in terms of π .

	Circle	Circumference
1.	Innermost Circle	6π
2.	Middle Circle	12π
3.	Outer Circle	18π

Complete the table.

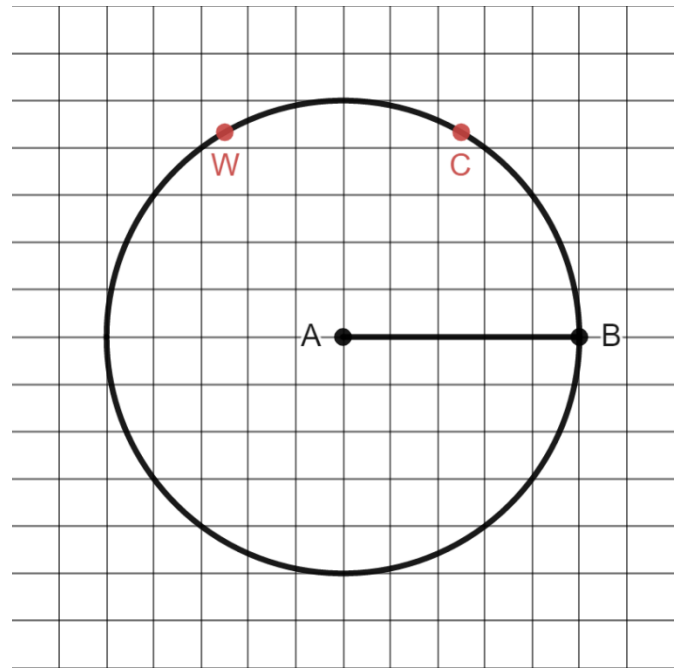
	Circle	Ratio of Circumference to Radius
4.	Innermost Circle	$\frac{6\pi}{3} = 2\pi$
5.	Middle Circle	$\frac{12\pi}{6} = 2\pi$
6.	Outer Circle	$\frac{18\pi}{9} = 2\pi$

7. What would be the ratio of the circumference to the radius for a circle whose center is W and whose radius has a measurement of 16?
- $\frac{32\pi}{16} = 2\pi$

Circle A is shown where $\overline{AB}$ is a radius.

8. Label point C on the circumference of circle A so that the resulting arc has an approximate length of 1 radius.

See grid.



9. Describe how you determined the location of point C .

Answers may vary. Example answer: I measured the length of the radius and measured that length on the circle.

10. Label point W on the circumference of circle A so that the resulting arc has an approximate length of 2 radii.

See grid.

Geometry
Unit 10
Lesson 4 Practice

Name Answer Key
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1. Circle C has a diameter of 18 inches (in.). Points T and S are on the circle and form the sector TCS . The central angle $\angle TCS$ in Circle C measures 120° . Determine the area of the sector. Express your answer in terms of π .

$$A = \frac{120}{360}(\pi)(9)^2 = 27\pi \text{ in}^2$$

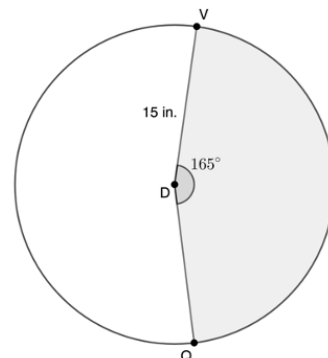
2. The area of a sector with a radius of 5 centimeters (cm) is 15.7 square cm. Calculate the approximate measure of the central angle of the sector in degrees.

$$15.7 = \frac{\theta}{360}(25)(\pi)$$

$$\theta \approx 71.96^\circ \text{ using } \pi \text{ button}$$

3. Circle D has radius $\overline{DV}$ that measures 15 inches (in.) with central angle $\angle VDQ$, where $m\angle VDQ = 165^\circ$. Find the area of the shaded sector. Round your answer to the nearest hundredth.

$$A = \frac{165}{360}(15)^2(\pi) \approx 323.98 \text{ in.}^2 \text{ using } \pi \text{ button}$$



4. Circle K has a radius of 7 inches (in.). Points R and Y are on the circle and form the sector RKY . The central angle $\angle RKY$ in Circle K measures 145° . Determine the area of the sector. Express your answer in terms of π .

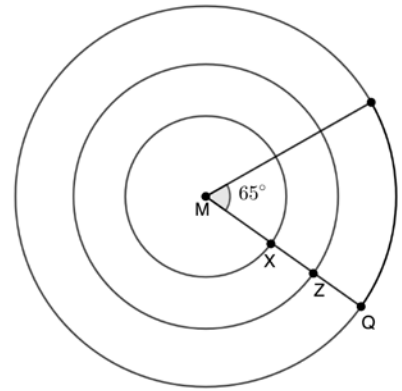
$$A = \frac{145}{360}(7)^2(\pi) \approx 19.74\pi \text{ in}^2$$

5. The area of a sector with a diameter of 22 feet (ft) is 27.44 square ft. Calculate the approximate measure of the central angle of the sector in degrees.

$$27.44 = \frac{\theta}{360}(11)^2(\pi)$$

$$\theta \approx 25.99^\circ \text{ using } \pi \text{ button}$$

For questions 6-8, concentric circles with center M are shown with $m\angle M = 65^\circ$. The smallest Circle M has a radius $\overline{MX}$ that measures 10 yards (yd). The largest Circle M has radius $\overline{MQ}$ and the middle Circle M has radius $\overline{MZ}$. Each circle has a radius that is 10 yd longer than the immediate prior circle.



6. What is the arc length of the circle with radius $\overline{MX}$?

Arc length = $2(\pi)(10)\left(\frac{65}{360}\right) \approx 11.34$ yds. using π button

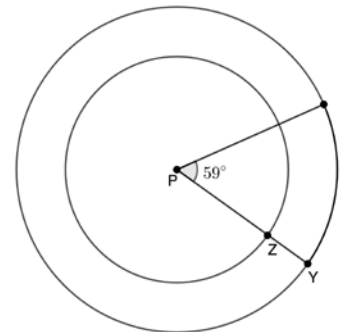
7. What is the arc length of the circle with radius $\overline{MZ}$?

Arc length = $2(\pi)(20)\left(\frac{65}{360}\right) \approx 22.69$ yds. using π button

8. What is the arc length of the circle with radius $\overline{MQ}$?

Arc length = $2(\pi)(30)\left(\frac{65}{360}\right) \approx 34.03$ yds. using π button

For questions 9-10, concentric circles with center P are shown with $m\angle P = 59^\circ$. The smallest Circle P has a radius $\overline{PZ}$ that measures 9 meters (m) and the largest Circle P has radius $\overline{PY}$. The radius of the largest Circle P has a radius 4 m larger than the smaller Circle P .



9. What is the arc length of the circle with radius $\overline{PZ}$?

Arc length = $2(\pi)(9)\left(\frac{59}{360}\right) \approx 9.27$ m. using π button

10. What is the arc length of the circle with radius $\overline{PY}$?

Arc length = $2(\pi)(13)\left(\frac{59}{360}\right) \approx 13.39$ m. using π button

Geometry
Unit 10
Lesson 5 Practice

Name Answer Key
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Period

1. Circle C has points X, N , and P on the circle with $m\angle NCX = 47.1^\circ$. If $m\widehat{NP} = 116.9^\circ$ and $\widehat{NP}$ does not intersect $\widehat{NX}$, what is $m\angle PCX$?

$$m\angle PCX = 196^\circ$$

2. Circle S has radii $\overline{ST}$ and $\overline{SR}$. Find the value of x if $m\angle RST = 135^\circ$ and $m\widehat{RT} = (6x + 9)^\circ$.

$$6x + 9 = 135$$

$$6x = 126$$

$$x = 21$$

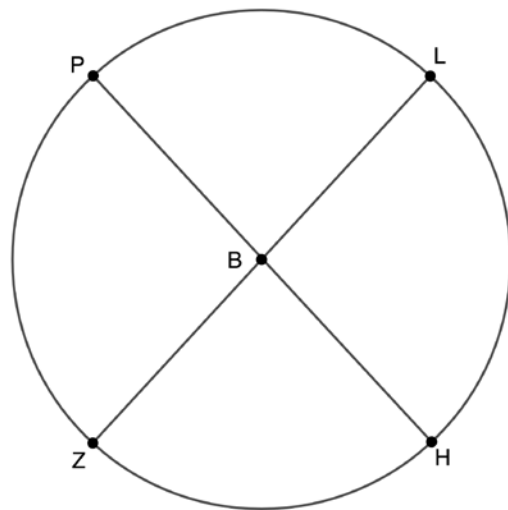
For questions 3-4, Circle B has diameters $\overline{PH}$ and $\overline{LZ}$ that both intersect at point B .

3. Given $m\widehat{HZ} = 72^\circ$, find the $m\widehat{PL}$.

$$m\widehat{PL} = 72^\circ$$

4. Given $m\widehat{HL} = 108^\circ$, find the $m\widehat{ZP}$.

$$m\widehat{ZP} = 108^\circ$$



For questions 5-6, consider the following context.

Lorenzo is creating a folding fan. He needs to determine the measure of the arc of the fan before he can cut the paper. He uses the expression $(6x + 110)^\circ$ as the measure of the central angle, $\angle ARQ$. For the $m\widehat{QA}$, he uses the expression $(24x + 38)^\circ$.

5. What is the value of x ?

$$6x + 110 = 24x + 38$$

$$72 = 18x$$

$$x = 4$$

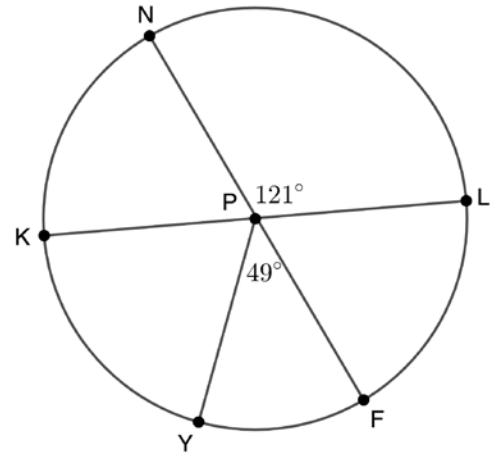
6. What is the degree measure of the arc of the fan, $\widehat{QA}$?

$$134^\circ$$

7. In Circle P , $m\angle NPL = 121^\circ$ and $m\angle FPY = 49^\circ$. Calculate the $m\widehat{NLY}$, given that P is the center and $\overline{NF}$ and $\overline{KL}$ are diameters.

$$\overline{LF} = 180 - 121 = 59^\circ$$

$$m\widehat{NLY} = 121 + 59 + 49 = 229^\circ$$



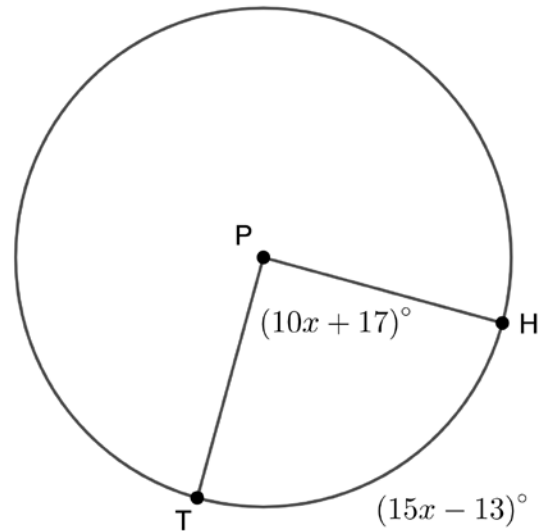
8. Circle S has points K, D , and Y on the circle.
 $m\angle YSD = 71^\circ$, $m\widehat{KY} = 132^\circ$, and $\widehat{KY}$ does not intersect $\widehat{YD}$.
 Find $m\angle KSD$.

$$360 - 71 - 132 = 157$$

$$m\angle KSD = 157^\circ$$

For questions 9-10, consider the following context.

A horse is tied to a post at point P and is grazing in a circular pattern. She decides to walk from her position at point H to the trough of water at point T along the circumference of the circular area she can roam. The measure of the central angle, $\angle TPH$, is $(10x + 17)^\circ$, and the measure of the arc that the horse walked, $\widehat{TH}$, is $(15x - 13)^\circ$.



9. Find the value of x .

$$10x + 17 = 15x - 13$$

$$30 = 5x$$

$$x = 6$$

10. Find the measure of the arc that the horse walked, $\widehat{TH}$.

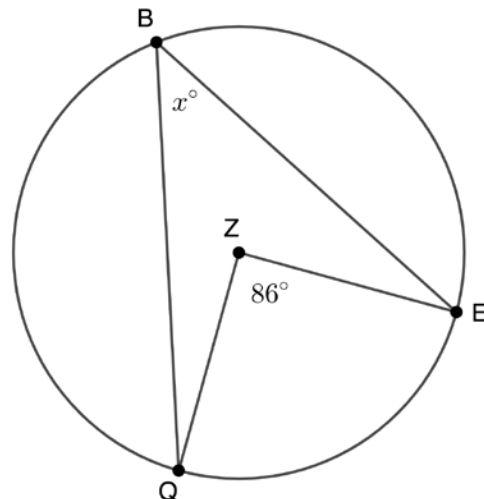
$$\widehat{TH} = 15(6) - 13 = 77^\circ$$

Geometry
Unit 10
Lesson 6 Practice

Name Answer Key
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1. Circle Z is shown where $\angle EZQ$ is a central angle, $\angle EBQ$ is inscribed in the circle, and $m\angle EZQ = 86^\circ$. Find $m\angle EBQ$.

$$m\angle EBQ = \frac{86}{2} = 43^\circ$$

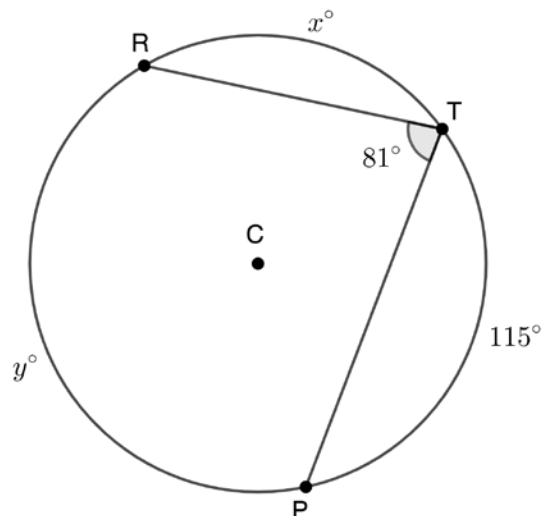


For questions 2-3, Circle C has points R , T , and P on the circle. $\angle RTP$ is inscribed in Circle C measuring 81° , and the

$$m\widehat{TP} = 115^\circ.$$

2. Find the $m\widehat{RP}$.

$$m\widehat{RP} = 2 \cdot 81 = 162^\circ$$



3. Find the $m\widehat{RT}$.

$$m\widehat{RT} = 360 - 115 - 162 = 83^\circ$$

For questions 4-5, points O , P , Q , and R lie on Circle S . $\angle PQR$ is inscribed in the circle, the $m\angle PQR = (5x + 22)^\circ$, and the $m\widehat{ROP} = 194^\circ$.

4. Find the $m\angle PQR$.

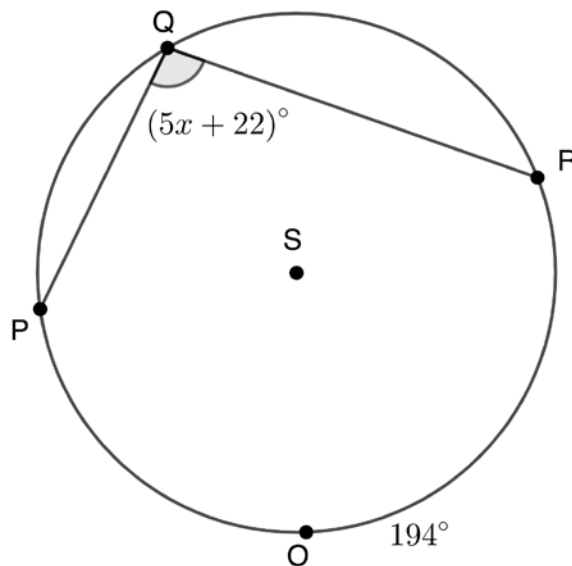
$$m\angle PQR = \frac{194}{2} = 97^\circ$$

5. Find the value of x .

$$5x + 22 = 97$$

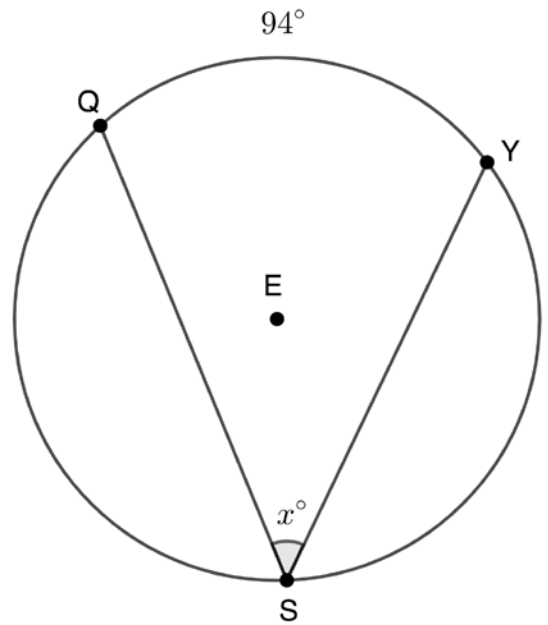
$$5x = 75$$

$$x = 15$$



6. Circle E is shown, where points Q , Y , and S are on the circle. $\angle QSY$ is inscribed in Circle E with $m\widehat{QY} = 94^\circ$. Find the value of x .

$$x = \frac{94}{2} = 47$$



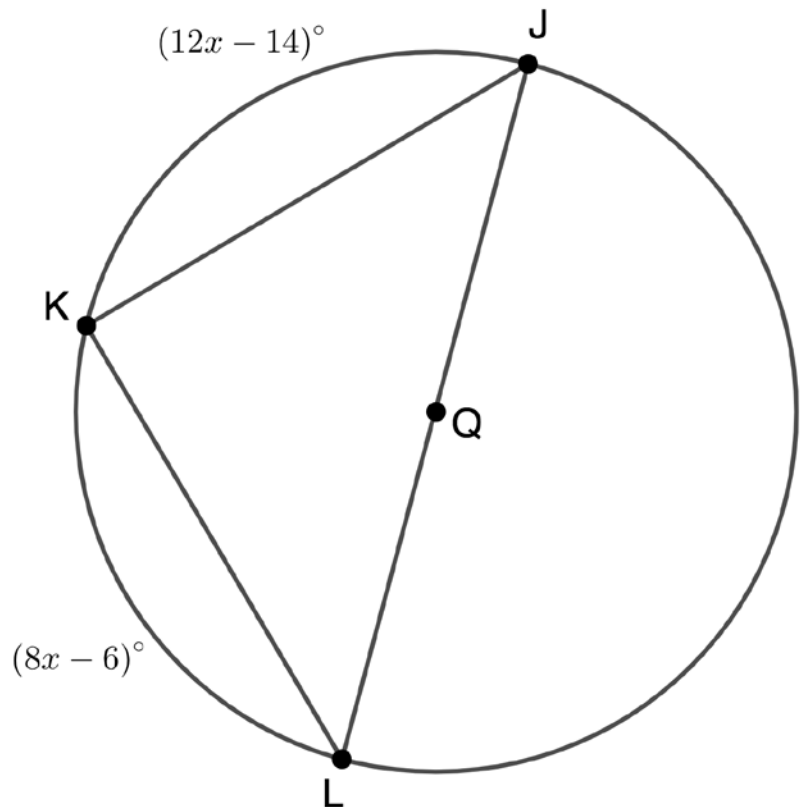
For questions 7-10, Circle Q is shown with inscribed angle $\angle LKJ$ and diameter $\overline{JL}$. The $m\widehat{KJ} = (12x - 14)^\circ$ and the $m\widehat{KL} = (8x - 6)^\circ$.

7. Find the value of x .
 $12x - 14 + 8x - 6 = 180$
 $20x - 20 = 180$
 $20x = 200$
 $x = 10$

8. Find the $m\angle KJL$.
 $\widehat{KL} = 8(10) - 6 = 74^\circ$
 $m\angle KJL = \frac{74}{2} = 37^\circ$

9. Find the $m\angle KLJ$.
 $\widehat{KJ} = 12(10) - 14 = 106^\circ$
 $m\angle KLJ = \frac{106}{2} = 53^\circ$

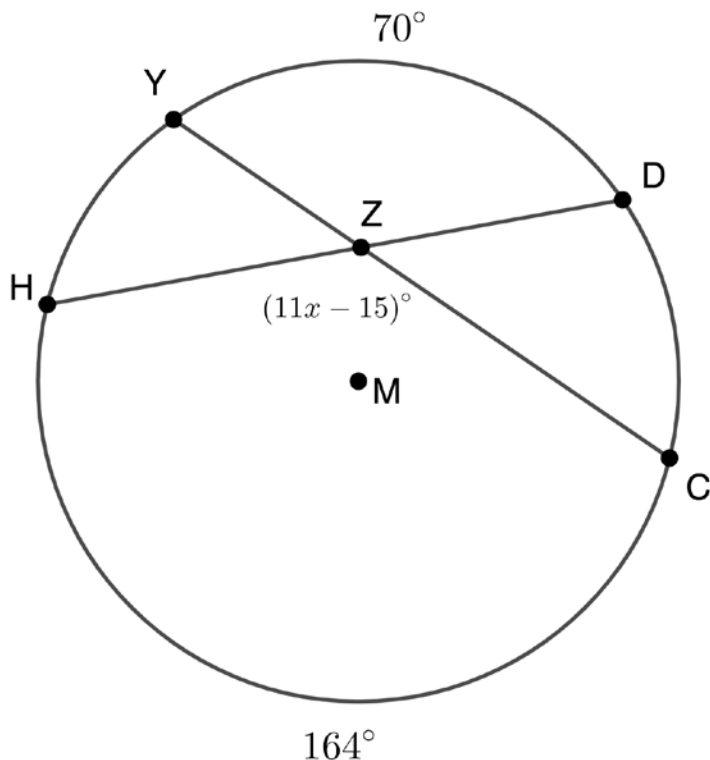
10. Find the $m\widehat{KJ}$.
 $m\widehat{KJ} = 12(10) - 14 = 106^\circ$



Geometry
Unit 10
Lesson 7 Practice

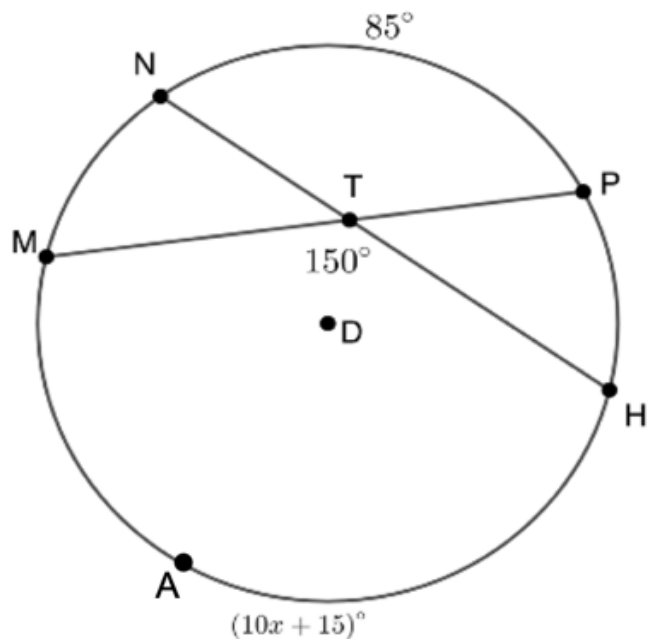
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For questions 1-3, Circle M is shown, where chords $\overline{YC}$ and $\overline{HD}$ intersect at point Z , $m\widehat{CH} = 164^\circ$, $m\widehat{YD} = 70^\circ$, and $m\angle HZC = (11x - 15)^\circ$.



- Find the $m\angle HZC$.
 $m\angle HZC = \frac{164+70}{2} = 117^\circ$
- Write an equation to find the value of x .
 $11x - 15 = 117$
- Find the value of x .
 $11x - 15 = 117$
 $11x = 132$
 $x = 12$

For questions 4-6, Circle D is shown, where chords $\overline{NH}$ and $\overline{MP}$ intersect at point T , $m\widehat{NP} = 85^\circ$, $m\widehat{MH} = (10x + 15)^\circ$, and $m\angle MTH = 150^\circ$.

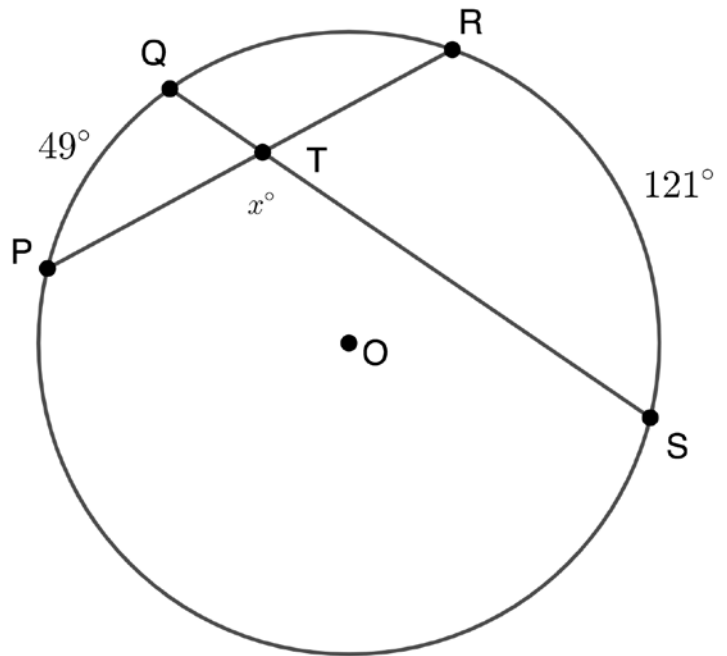


- Write an equation to find the value of x .
 $\frac{(10x+15)+85}{2} = 150$
- Find the value of x .
 $\frac{(10x+15)+85}{2} = 150$
 $10x + 100 = 300$
 $10x = 200$
 $x = 20$
- Find the $m\widehat{MH}$.
 $m\widehat{MH} = 10(20) + 15 = 215^\circ$

7. Circle O is shown, where chords $\overline{QS}$ and $\overline{RP}$ intersect at point T . Find the value of x for $\angle PTS$, given $m\widehat{RS} = 121^\circ$ and $m\widehat{PQ} = 49^\circ$.

$$m\angle RTS = \frac{121+49}{2} = 85^\circ$$

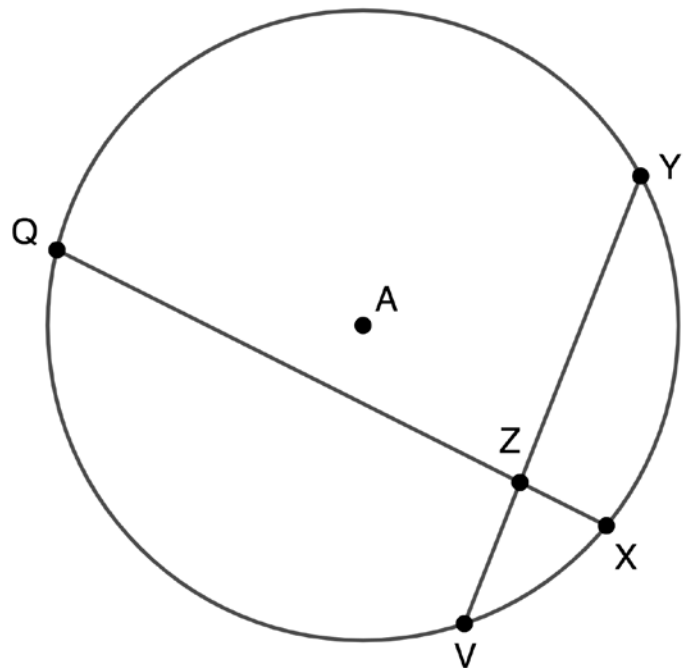
$$x = 180 - 85 = 95$$



For questions 8-10, Circle A is shown, where chords $\overline{QX}$ and $\overline{YV}$ intersect at point Z .

8. If $m\widehat{QY} = 137^\circ$ and $m\widehat{VX} = 35^\circ$, what is the measure of $\angle QZY$?

$$m\angle QZY = \frac{137+35}{2} = 86^\circ$$



9. If $m\widehat{QV} = 141^\circ$ and $m\widehat{YX} = 47^\circ$, what is the measure of $\angle QZV$?

$$m\angle QZV = \frac{141+47}{2} = 94^\circ$$

10. If $m\widehat{VX} = 41^\circ$ and $m\angle QZY = 84^\circ$, what is the measure of $\widehat{QY}$?

$$84 = \frac{41+m\widehat{QY}}{2}$$

$$168 = 41 + m\widehat{QY}$$

$$m\widehat{QY} = 127^\circ$$

Geometry
Unit 10
Lesson 8 Practice

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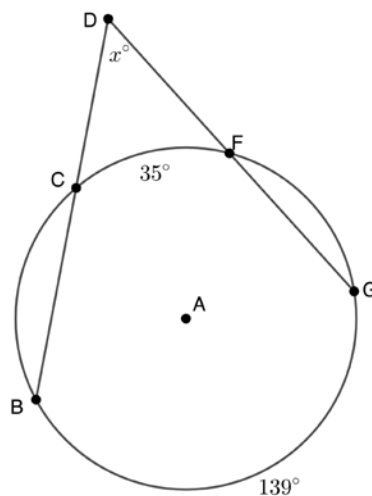
For questions 1-2, Circle A has points B , C , F , and G on the circle. Secant lines $\overline{BD}$ and $\overline{GD}$ intersect at point D outside the circle. The $m\widehat{BG} = 139^\circ$, $m\widehat{CF} = 35^\circ$, and $m\angle CDF = x^\circ$.

1. Write a formula that can be used to find the value of x .

$$x = \frac{139-35}{2}$$

2. Find the value of x .

$$x = \frac{139-35}{2} = 52$$



For questions 3-5, Circle K has points N , H , L , and M on the circle. Secant lines $\overline{NJ}$ and $\overline{MJ}$ intersect at point J outside the circle. The $m\widehat{LH} = (x + 12)^\circ$, $m\widehat{MN} = (3x - 28)^\circ$, and $m\angle HJL = 28^\circ$.

3. Find the value of x .

$$\frac{(3x-28)-(x+12)}{2} = 28$$

$$2x - 40 = 56$$

$$2x = 96$$

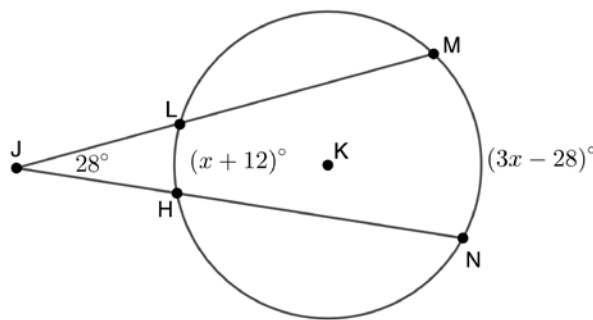
$$x = 48$$

4. Find the $m\widehat{MN}$.

$$m\widehat{MN} = 3(48) - 28 = 116^\circ$$

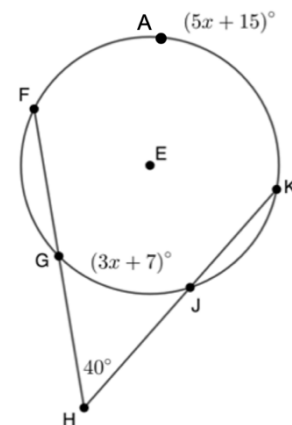
5. Find the $m\widehat{LH}$.

$$m\widehat{LH} = 48 + 12 = 60^\circ$$



Not drawn to scale

For questions 6-8, Circle E has points F , G , J , and K on the circle. Secant lines $\overline{FH}$ and $\overline{KH}$ intersect at point H outside the circle. The $m\widehat{FAK} = (5x + 15)^\circ$, $m\widehat{GJ} = (3x + 7)^\circ$, and $m\angle GHJ = 40^\circ$.



Not drawn to scale

6. Find the value of x .

$$\frac{(5x+15)-(3x+7)}{2} = 40$$

$$2x + 8 = 80$$

$$2x = 72$$

$$x = 36$$

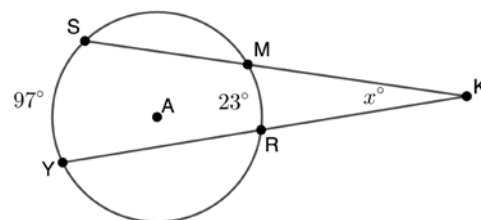
7. Find the $m\widehat{GJ}$.

$$m\widehat{GJ} = 3(36) + 7 = 115^\circ$$

8. Find the $m\widehat{FAK}$.

$$m\widehat{FAK} = 5(36) + 15 = 195^\circ$$

For questions 9-10, Circle A has points S , M , Y , and R on the circle. Secant lines $\overline{SK}$ and $\overline{YK}$ intersect at point K outside the circle. The $m\widehat{SY} = 97^\circ$, $m\widehat{MR} = 23^\circ$, and $m\angle MKR = x^\circ$.



Not drawn to scale

9. Write a formula that can be used to find the value of x .

$$\frac{97-23}{2} = x$$

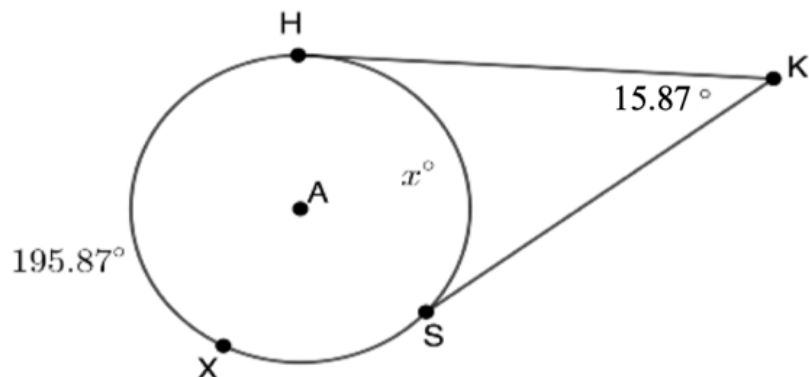
10. Find the value of x .

$$x = \frac{97-23}{2} = 37$$

Geometry
Unit 10
Lesson 9 Practice

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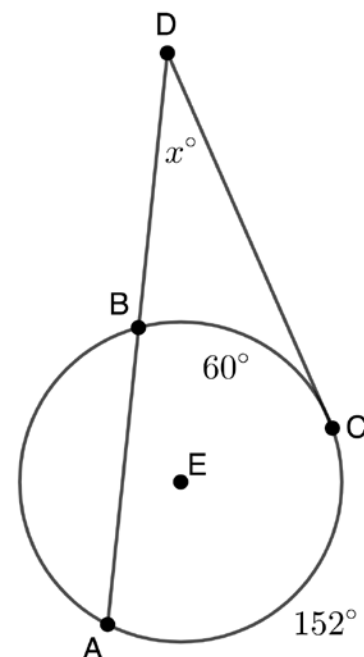
For questions 1-2, Circle A has points H, X , and S on the circle. Tangents $\overline{HK}$ and $\overline{SK}$ intersect at point K outside the circle. The $m\widehat{HXS} = 195.87^\circ$, $m\widehat{HS} = x^\circ$, and $m\angle HKS = 15.87^\circ$.



- Write an equation that can be used to find the value of x .
 $360 - 195.87 = x$ or $\frac{195.87 - x}{2} = 15.87$

- Find the value of x .
 $x = 360 - 195.87 = 164.13$

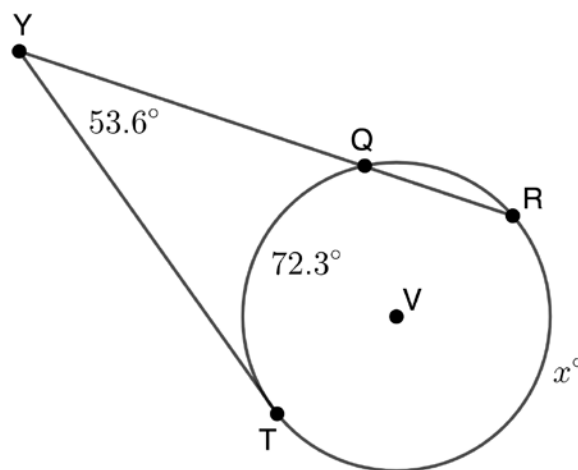
For questions 3-4, Circle E has points A, B , and C on the circle. Tangent $\overline{CD}$ and secant $\overline{AD}$ intersect at point D outside the circle. The $m\widehat{AC} = 152^\circ$, $m\widehat{BC} = 60^\circ$, and $m\angle BDC = x^\circ$.



- Write an equation that can be used to find the value of x .
 $x = \frac{152 - 60}{2}$

- Find the value of x .
 $x = \frac{152 - 60}{2} = 46$

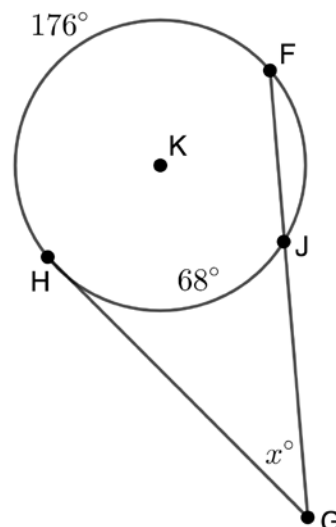
For questions 5-6, Circle V has points T, Q , and R on the circle. Tangent $\overline{TY}$ and secant $\overline{RY}$ intersect at point Y outside the circle. The $m\widehat{TR} = x^\circ$, $m\widehat{TQ} = 72.3^\circ$, and $m\angle TYQ = 53.6^\circ$.



- Write an equation that can be used to find the value of x .
 $\frac{x - 72.3}{2} = 53.6$

- Find the value of x .
 $\frac{x - 72.3}{2} = 53.6$
 $x - 72.3 = 107.2$
 $x = 179.5$

7. Circle K has points F, H , and J on the circle. Tangent $\overline{HG}$ and secant $\overline{FG}$ intersect at point G outside the circle. The $m\widehat{HF} = 176^\circ$, $m\widehat{HJ} = 68^\circ$, and $m\angle JGH = x^\circ$. What is the $m\angle JGH$?
- $$m\angle JGH = \frac{176 - 68}{2} = 54^\circ$$

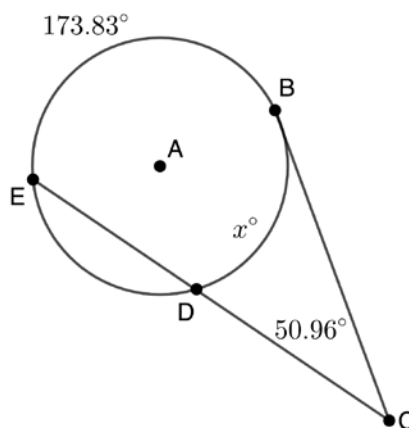


8. Circle A has points B, D , and E on the circle. Tangent $\overline{BC}$ and secant $\overline{EC}$ intersect at point C outside the circle. The $m\widehat{EB} = 173.83^\circ$, $m\widehat{DB} = x^\circ$, and $m\angle BCD = 50.96^\circ$. What is the value of x ?

$$\frac{173.83 - x}{2} = 50.96$$

$$173.83 - x = 101.92$$

$$x = 71.91$$



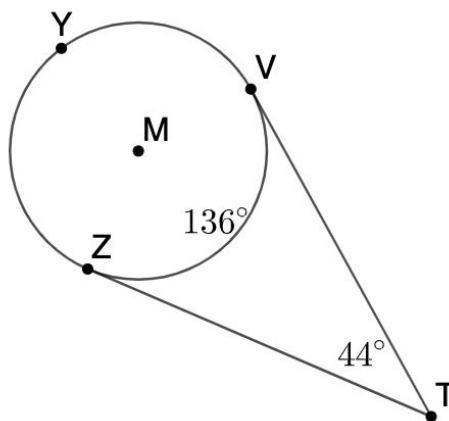
9. Circle M has points V, Y , and Z on the circle. Tangents $\overline{ZT}$ and $\overline{VT}$ intersect at point T outside the circle. $m\widehat{VZ} = 136^\circ$, and $m\angle VTZ = 44^\circ$. What is the $m\widehat{VYZ}$?

$$\frac{x - 136}{2} = 44$$

$$x - 136 = 88$$

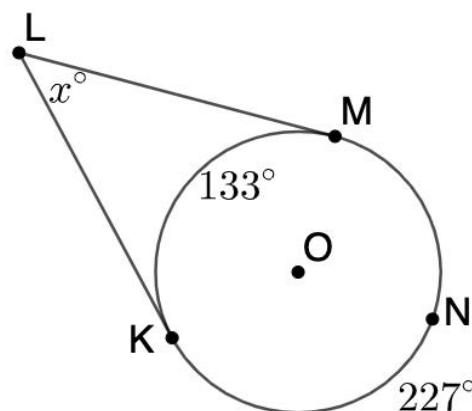
$$x = 224$$

$$m\widehat{VYZ} = 224^\circ$$

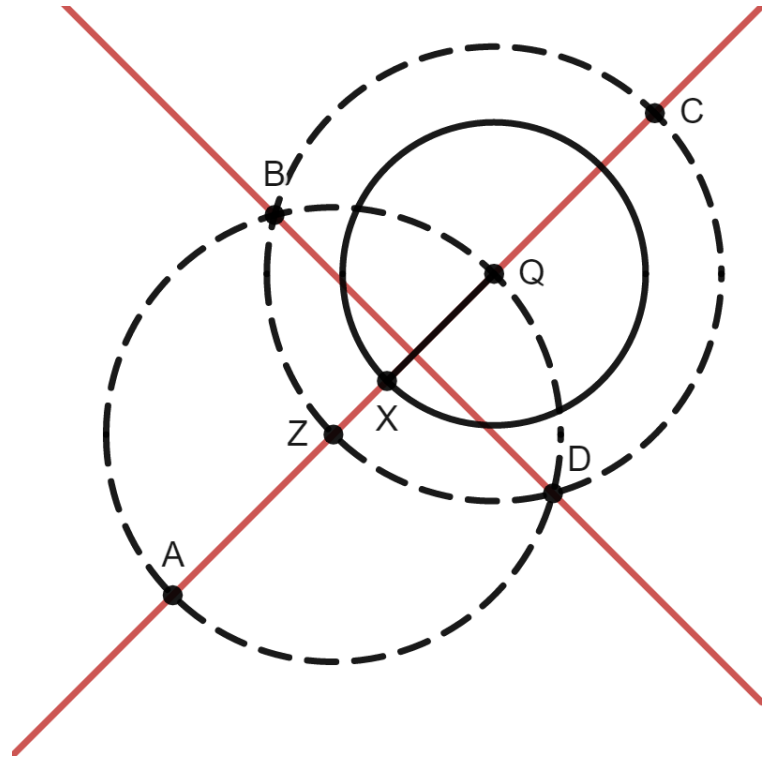


10. Circle O has points K, M , and N on the circle. Tangents $\overline{KL}$ and $\overline{ML}$ intersect at point L outside the circle. The $m\widehat{MNK} = 227^\circ$, $m\widehat{MK} = 133^\circ$, and $m\angle KLM = x^\circ$. What is the $m\angle KLM$?

$$m\angle KLM = \frac{227 - 133}{2} = 47^\circ$$



Circle Q is shown with radius $\overline{QX}$. The two dotted circles were constructed using a compass so that they are congruent to each other. Points C , B , D , and Z are on the dotted circle Q and points B , Q , D , and A on the dotted circle Z .



1. Select all of the line segments that are congruent to each other.

- ☐ $\overline{BC}$
☒ $\overline{BZ}$
☒ $\overline{BQ}$
☒ $\overline{QD}$
☒ $\overline{QC}$
☒ $\overline{QZ}$
☐ $\overline{QX}$

2. Justify why the line segments chosen in question 1 are congruent.

Answers may vary. Sample response: All line segments are the radii of both congruent circles.

3. Select all of the line segments that are congruent to each other.

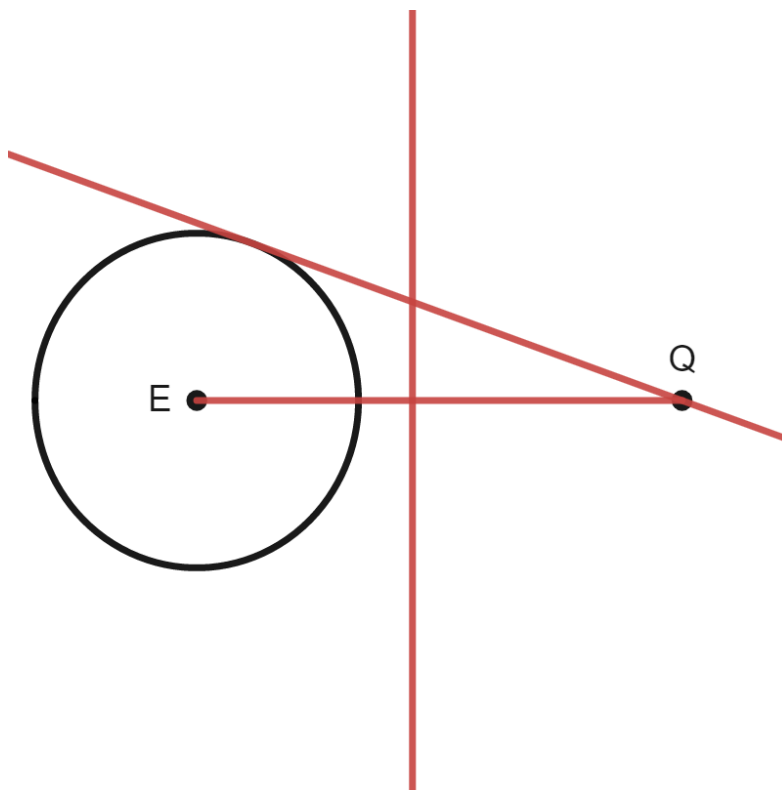
- ☐ $\overline{BD}$
☐ $\overline{AB}$
☒ $\overline{BZ}$
☐ $\overline{AD}$
☒ $\overline{AZ}$
☒ $\overline{ZD}$
☒ $\overline{ZQ}$
☐ $\overline{ZX}$

4. Justify why the line segments chosen in question 3 are congruent.

Answers may vary. Sample response: The line segments are radii in Circle Z .

5. Draw $\overleftrightarrow{AC}$ on the diagram.
See diagram.
6. Draw $\overleftrightarrow{BD}$ on the diagram.
See diagram.
7. Describe how $\overleftrightarrow{AC}$ and $\overleftrightarrow{BD}$ are related.
 $\overleftrightarrow{AC} \perp \overleftrightarrow{BD}$
8. Justify whether $\overleftrightarrow{BD}$ is tangent to circle Q .
 $\overleftrightarrow{BD}$ is not tangent to Circle Q . It meets Circle Q twice. Need Point X as midpoint of $\overline{ZQ}$ by separating Circle Z and Circle Q some more.

Circle E and point Q are shown.



9. Construct the perpendicular bisector of $\overline{EQ}$.
See diagram.
10. Construct a line that is tangent to Circle E and goes through point Q .
See diagram.